

Montgomery County Housing EDA

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2026-04-20

Libraries

```
library(tidyverse)
library(lubridate)
library(scales)
```

Load Zillow City Data

```
housing_city <- read.csv(
  "~/Downloads/City_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv",
  check.names = FALSE
)
```

Filter Montgomery County Cities

```
# Filter only Montgomery County, Maryland cities
moco_cities <- housing_city %>%
  filter(StateName == "MD", CountyName == "Montgomery County")

# Display city names
unique(moco_cities$RegionName)
```

```
## [1] "Silver Spring"      "Rockville"          "Gaithersburg"
## [4] "Germantown"         "Bethesda"           "Potomac"
## [7] "Montgomery Village" "Takoma Park"        "Chevy Chase"
## [10] "North Potomac"     "Olney"              "Derwood"
## [13] "Clarksburg"        "Burtonsville"       "Kensington"
## [16] "Damascus"          "Boyd's"             "Brookeville"
## [19] "Darnestown"        "Poolesville"        "Sandy Spring"
## [22] "Dickerson"         "Cabin John"         "Ashton"
## [25] "Chevy Chase View"  "Garrett Park"       "Laytonsville"
## [28] "Washington Grove"  "Spencerville"       "Brinklow"
## [31] "Glen Echo"
```

Initial Data Inspection

```
# Inspect structure of the dataset  
glimpse(moco_cities)
```

```

## Rows: 31
## Columns: 323
## $ RegionID      <int> 7081, 33714, 45383, 31732, 37406, 47210, 19445, 47942, 39...
## $ SizeRank      <int> 79, 421, 425, 587, 652, 1289, 1883, 2438, 2519, 2520, 271...
## $ RegionName    <chr> "Silver Spring", "Rockville", "Gaithersburg", "Germantown...
## $ RegionType    <chr> "city", "city", "city", "city", "city", "city", "city", "...
## $ StateName     <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD...
## $ State         <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD...
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##	[46]	"2003-02-28"	"2003-03-31"	"2003-04-30"	"2003-05-31"	"2003-06-30"
##	[51]	"2003-07-31"	"2003-08-31"	"2003-09-30"	"2003-10-31"	"2003-11-30"
##	[56]	"2003-12-31"	"2004-01-31"	"2004-02-29"	"2004-03-31"	"2004-04-30"
##	[61]	"2004-05-31"	"2004-06-30"	"2004-07-31"	"2004-08-31"	"2004-09-30"
##	[66]	"2004-10-31"	"2004-11-30"	"2004-12-31"	"2005-01-31"	"2005-02-28"
##	[71]	"2005-03-31"	"2005-04-30"	"2005-05-31"	"2005-06-30"	"2005-07-31"
##	[76]	"2005-08-31"	"2005-09-30"	"2005-10-31"	"2005-11-30"	"2005-12-31"
##	[81]	"2006-01-31"	"2006-02-28"	"2006-03-31"	"2006-04-30"	"2006-05-31"
##	[86]	"2006-06-30"	"2006-07-31"	"2006-08-31"	"2006-09-30"	"2006-10-31"
##	[91]	"2006-11-30"	"2006-12-31"	"2007-01-31"	"2007-02-28"	"2007-03-31"
##	[96]	"2007-04-30"	"2007-05-31"	"2007-06-30"	"2007-07-31"	"2007-08-31"
##	[101]	"2007-09-30"	"2007-10-31"	"2007-11-30"	"2007-12-31"	"2008-01-31"
##	[106]	"2008-02-29"	"2008-03-31"	"2008-04-30"	"2008-05-31"	"2008-06-30"
##	[111]	"2008-07-31"	"2008-08-31"	"2008-09-30"	"2008-10-31"	"2008-11-30"
##	[116]	"2008-12-31"	"2009-01-31"	"2009-02-28"	"2009-03-31"	"2009-04-30"
##	[121]	"2009-05-31"	"2009-06-30"	"2009-07-31"	"2009-08-31"	"2009-09-30"
##	[126]	"2009-10-31"	"2009-11-30"	"2009-12-31"	"2010-01-31"	"2010-02-28"
##	[131]	"2010-03-31"	"2010-04-30"	"2010-05-31"	"2010-06-30"	"2010-07-31"
##	[136]	"2010-08-31"	"2010-09-30"	"2010-10-31"	"2010-11-30"	"2010-12-31"
##	[141]	"2011-01-31"	"2011-02-28"	"2011-03-31"	"2011-04-30"	"2011-05-31"
##	[146]	"2011-06-30"	"2011-07-31"	"2011-08-31"	"2011-09-30"	"2011-10-31"
##	[151]	"2011-11-30"	"2011-12-31"	"2012-01-31"	"2012-02-29"	"2012-03-31"
##	[156]	"2012-04-30"	"2012-05-31"	"2012-06-30"	"2012-07-31"	"2012-08-31"
##	[161]	"2012-09-30"	"2012-10-31"	"2012-11-30"	"2012-12-31"	"2013-01-31"
##	[166]	"2013-02-28"	"2013-03-31"	"2013-04-30"	"2013-05-31"	"2013-06-30"
##	[171]	"2013-07-31"	"2013-08-31"	"2013-09-30"	"2013-10-31"	"2013-11-30"
##	[176]	"2013-12-31"	"2014-01-31"	"2014-02-28"	"2014-03-31"	"2014-04-30"
##	[181]	"2014-05-31"	"2014-06-30"	"2014-07-31"	"2014-08-31"	"2014-09-30"
##	[186]	"2014-10-31"	"2014-11-30"	"2014-12-31"	"2015-01-31"	"2015-02-28"
##	[191]	"2015-03-31"	"2015-04-30"	"2015-05-31"	"2015-06-30"	"2015-07-31"
##	[196]	"2015-08-31"	"2015-09-30"	"2015-10-31"	"2015-11-30"	"2015-12-31"
##	[201]	"2016-01-31"	"2016-02-29"	"2016-03-31"	"2016-04-30"	"2016-05-31"
##	[206]	"2016-06-30"	"2016-07-31"	"2016-08-31"	"2016-09-30"	"2016-10-31"
##	[211]	"2016-11-30"	"2016-12-31"	"2017-01-31"	"2017-02-28"	"2017-03-31"
##	[216]	"2017-04-30"	"2017-05-31"	"2017-06-30"	"2017-07-31"	"2017-08-31"
##	[221]	"2017-09-30"	"2017-10-31"	"2017-11-30"	"2017-12-31"	"2018-01-31"
##	[226]	"2018-02-28"	"2018-03-31"	"2018-04-30"	"2018-05-31"	"2018-06-30"
##	[231]	"2018-07-31"	"2018-08-31"	"2018-09-30"	"2018-10-31"	"2018-11-30"
##	[236]	"2018-12-31"	"2019-01-31"	"2019-02-28"	"2019-03-31"	"2019-04-30"
##	[241]	"2019-05-31"	"2019-06-30"	"2019-07-31"	"2019-08-31"	"2019-09-30"
##	[246]	"2019-10-31"	"2019-11-30"	"2019-12-31"	"2020-01-31"	"2020-02-29"
##	[251]	"2020-03-31"	"2020-04-30"	"2020-05-31"	"2020-06-30"	"2020-07-31"
##	[256]	"2020-08-31"	"2020-09-30"	"2020-10-31"	"2020-11-30"	"2020-12-31"

```
## [261] "2021-01-31" "2021-02-28" "2021-03-31" "2021-04-30" "2021-05-31"
## [266] "2021-06-30" "2021-07-31" "2021-08-31" "2021-09-30" "2021-10-31"
## [271] "2021-11-30" "2021-12-31" "2022-01-31" "2022-02-28" "2022-03-31"
## [276] "2022-04-30" "2022-05-31" "2022-06-30" "2022-07-31" "2022-08-31"
## [281] "2022-09-30" "2022-10-31" "2022-11-30" "2022-12-31" "2023-01-31"
## [286] "2023-02-28" "2023-03-31" "2023-04-30" "2023-05-31" "2023-06-30"
## [291] "2023-07-31" "2023-08-31" "2023-09-30" "2023-10-31" "2023-11-30"
## [296] "2023-12-31" "2024-01-31" "2024-02-29" "2024-03-31" "2024-04-30"
## [301] "2024-05-31" "2024-06-30" "2024-07-31" "2024-08-31" "2024-09-30"
## [306] "2024-10-31" "2024-11-30" "2024-12-31" "2025-01-31" "2025-02-28"
## [311] "2025-03-31" "2025-04-30" "2025-05-31" "2025-06-30" "2025-07-31"
## [316] "2025-08-31" "2025-09-30" "2025-10-31" "2025-11-30" "2025-12-31"
## [321] "2026-01-31" "2026-02-28" "2026-03-31"
```

Reshape Zillow Data

```
# Convert Zillow data from wide format to long format
# This makes time series analysis and plotting easier
moco_long <- moco_cities %>%
  pivot_longer(
    cols = matches("^\\d{4}-\\d{2}-\\d{2}$|^X\\d{4}\\.\\d{2}\\.\\d{2}$"),
    names_to = "Date",
    values_to = "ZHVI"
  ) %>%
  mutate(
    Date = str_remove(Date, "^X"),
    Date = str_replace_all(Date, "\\.", "-"),
    Date = as.Date(Date)
  )
```

Post-Cleaning Inspection

```
# Inspect cleaned data
glimpse(moco_long)
```

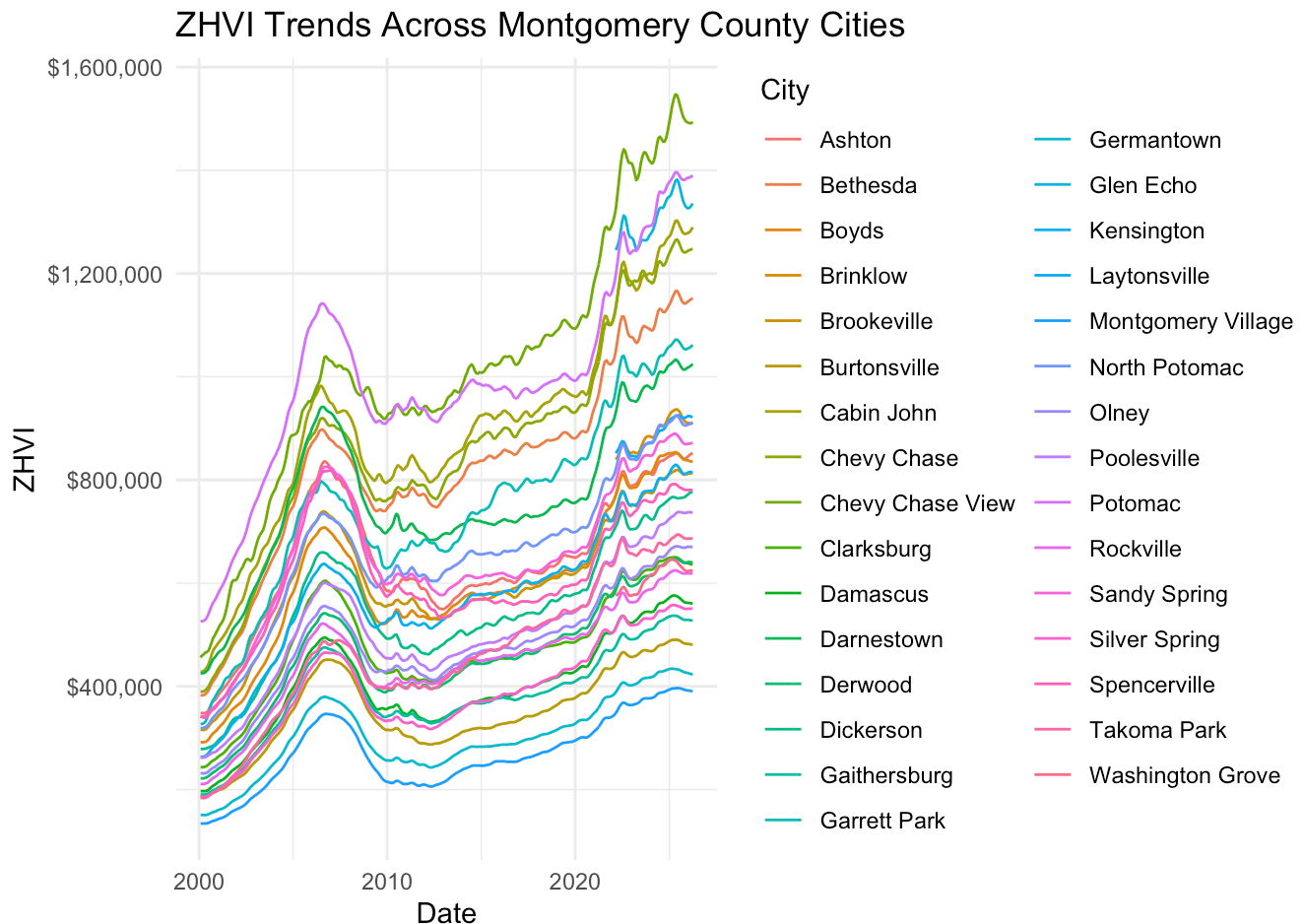
```
## Rows: 9,765
## Columns: 10
## $ RegionID    <int> 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081,...
## $ SizeRank    <int> 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79,...
## $ RegionName  <chr> "Silver Spring", "Silver Spring", "Silver Spring", "Silver ...
## $ RegionType  <chr> "city", "city", "city", "city", "city", "city", "city", "ci...
## $ StateName   <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD",...
## $ State       <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD",...
## $ Metro       <chr> "Washington-Arlington-Alexandria, DC-VA-MD-WV", "Washington...
## $ CountyName  <chr> "Montgomery County", "Montgomery County", "Montgomery Count...
## $ Date        <date> 2000-01-31, 2000-02-29, 2000-03-31, 2000-04-30, 2000-05-31...
## $ ZHVI        <dbl> 185818.2, 185642.1, 185688.0, 185978.9, 186870.6, 188282.9,...
```

```
# Summary statistics for ZHVI
summary(moco_long$ZHVI)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 134181  414426   564403   599572  756806 1547109     1060
```

ZHVI Trends Across Montgomery County Cities

```
ggplot(moco_long, aes(x = Date, y = ZHVI, color = RegionName)) +
  geom_line() +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "ZHVI Trends Across Montgomery County Cities",
    x = "Date",
    y = "ZHVI",
    color = "City"
  ) +
  theme_minimal()
```



Latest Home Prices

```
# Find the most recent Zillow Home Value Index for each city
latest_prices <- moco_long %>%
  group_by(RegionName) %>%
  filter(Date == max(Date, na.rm = TRUE)) %>%
  ungroup() %>%
  select(RegionName, Date, ZHVI) %>%
  arrange(desc(ZHVI))

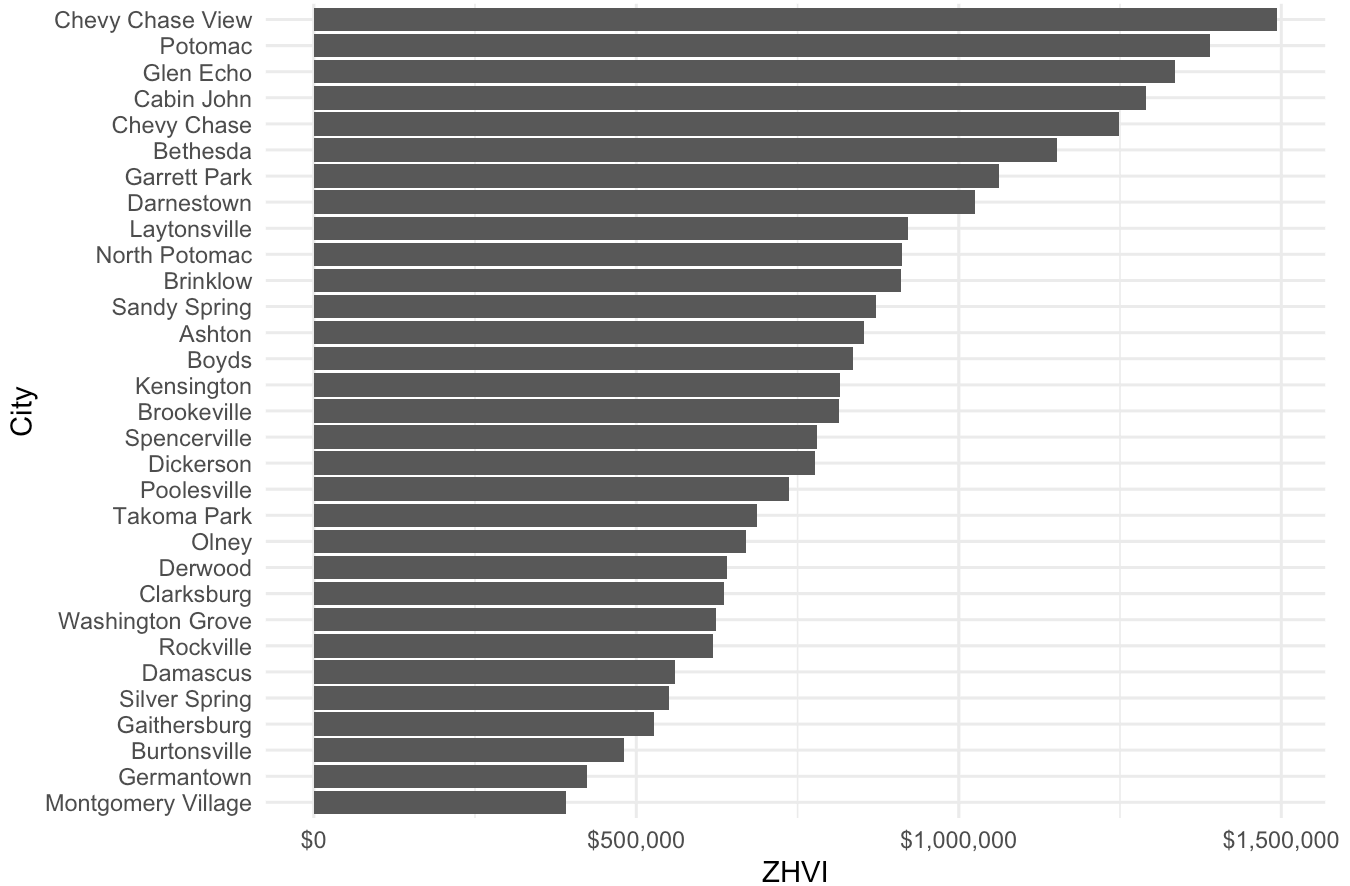
latest_prices
```

```
## # A tibble: 31 × 3
##   RegionName      Date      ZHVI
##   <chr>         <date>    <dbl>
## 1 Chevy Chase View 2026-03-31 1493700.
## 2 Potomac         2026-03-31 1390212.
## 3 Glen Echo       2026-03-31 1335696.
## 4 Cabin John      2026-03-31 1289768.
## 5 Chevy Chase     2026-03-31 1247761.
## 6 Bethesda        2026-03-31 1152219.
## 7 Garrett Park    2026-03-31 1061769.
## 8 Darnestown      2026-03-31 1024744.
## 9 Laytonsville    2026-03-31  922014.
## 10 North Potomac  2026-03-31  911608.
## # i 21 more rows
```

Latest ZHVI by City

```
ggplot(latest_prices, aes(x = reorder(RegionName, ZHVI), y = ZHVI)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "Most Recent ZHVI by Montgomery County City",
    x = "City",
    y = "ZHVI"
  ) +
  theme_minimal()
```

Most Recent ZHVI by Montgomery County City



Long-Term Growth Analysis

```
# Calculate long-term growth for each city
growth_summary <- moco_long %>%
  filter(!is.na(ZHVI)) %>%
  group_by(RegionName) %>%
  summarise(
    start_date = min(Date),
    end_date = max(Date),
    start_price = ZHVI[which.min(Date)],
    end_price = ZHVI[which.max(Date)],
    growth_dollars = end_price - start_price,
    growth_percent = (end_price - start_price) / start_price
  ) %>%
  arrange(desc(growth_percent))

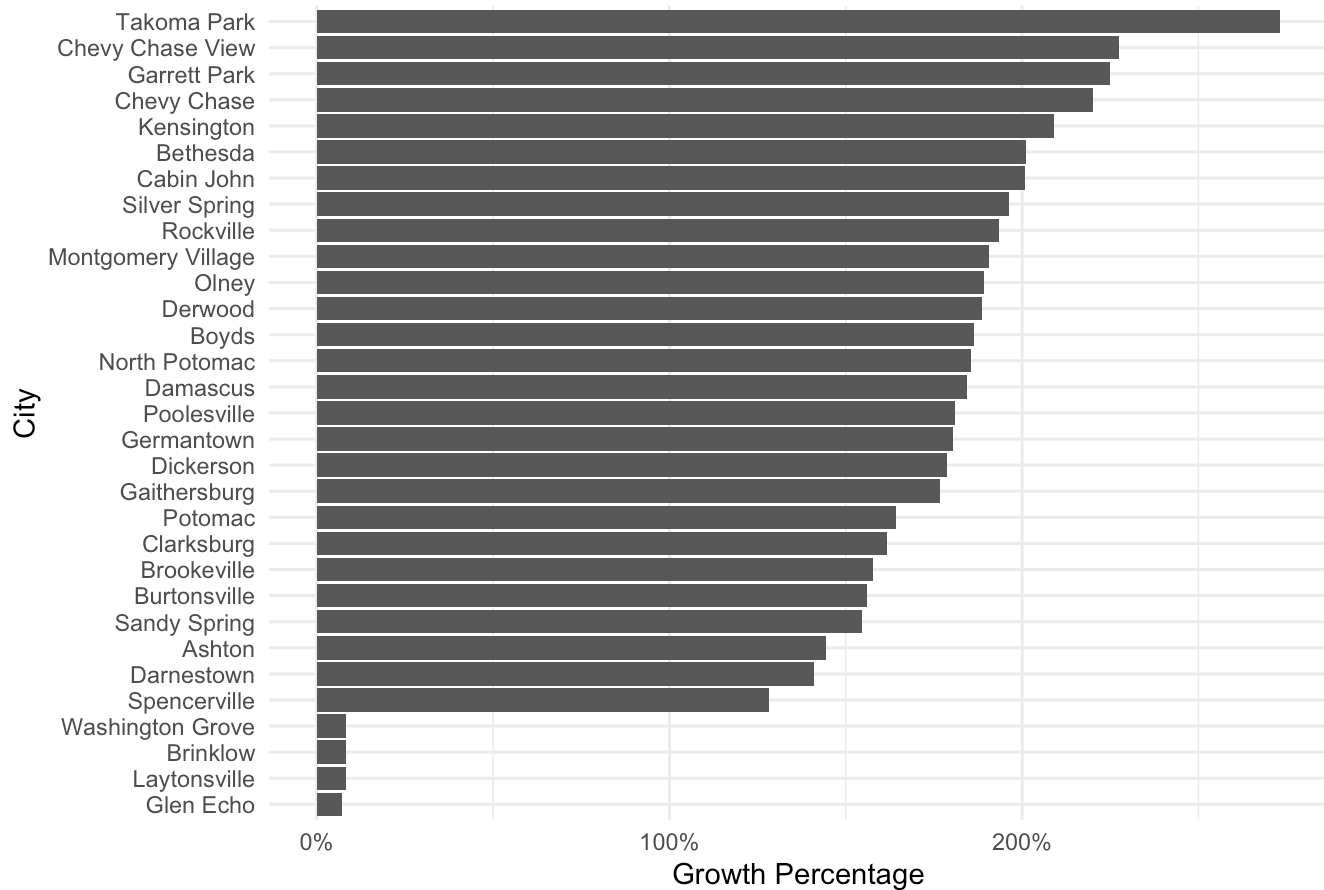
growth_summary
```

```
## # A tibble: 31 × 7
##   RegionName      start_date end_date   start_price end_price growth_dollars
##   <chr>          <date>    <date>         <dbl>      <dbl>        <dbl>
## 1 Takoma Park    2000-01-31 2026-03-31    183931.    686399.      502468.
## 2 Chevy Chase View 2000-01-31 2026-03-31    456026.   1493700.     1037674.
## 3 Garrett Park    2000-01-31 2026-03-31    326681.   1061769.      735088.
## 4 Chevy Chase     2000-01-31 2026-03-31    389743.   1247761.      858019.
## 5 Kensington      2000-01-31 2026-03-31    263815.    815355.      551540.
## 6 Bethesda        2000-01-31 2026-03-31    382718.   1152219.      769501.
## 7 Cabin John       2000-01-31 2026-03-31    428641.   1289768.      861127.
## 8 Silver Spring    2000-01-31 2026-03-31    185818.    550712.      364894.
## 9 Rockville        2000-01-31 2026-03-31    210841.    618892.      408051.
## 10 Montgomery Village 2000-01-31 2026-03-31    134430.    390564.      256134.
## # i 21 more rows
## # i 1 more variable: growth_percent <dbl>
```

Long-Term Growth by City

```
ggplot(growth_summary, aes(x = reorder(RegionName, growth_percent), y = growth_percent))
+
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Long-Term ZHVI Growth by Montgomery County City",
    x = "City",
    y = "Growth Percentage"
  ) +
  theme_minimal()
```

Long-Term ZHVI Growth by Montgomery County City



Volatility Analysis

```
# Calculate housing market volatility using monthly percent changes
volatility_summary <- moco_long %>%
  filter(!is.na(ZHVI)) %>%
  arrange(RegionName, Date) %>%
  group_by(RegionName) %>%
  mutate(
    monthly_change = (ZHVI - lag(ZHVI)) / lag(ZHVI)
  ) %>%
  summarise(
    avg_monthly_change = mean(monthly_change, na.rm = TRUE),
    volatility = sd(monthly_change, na.rm = TRUE)
  ) %>%
  arrange(desc(volatility))

volatility_summary
```



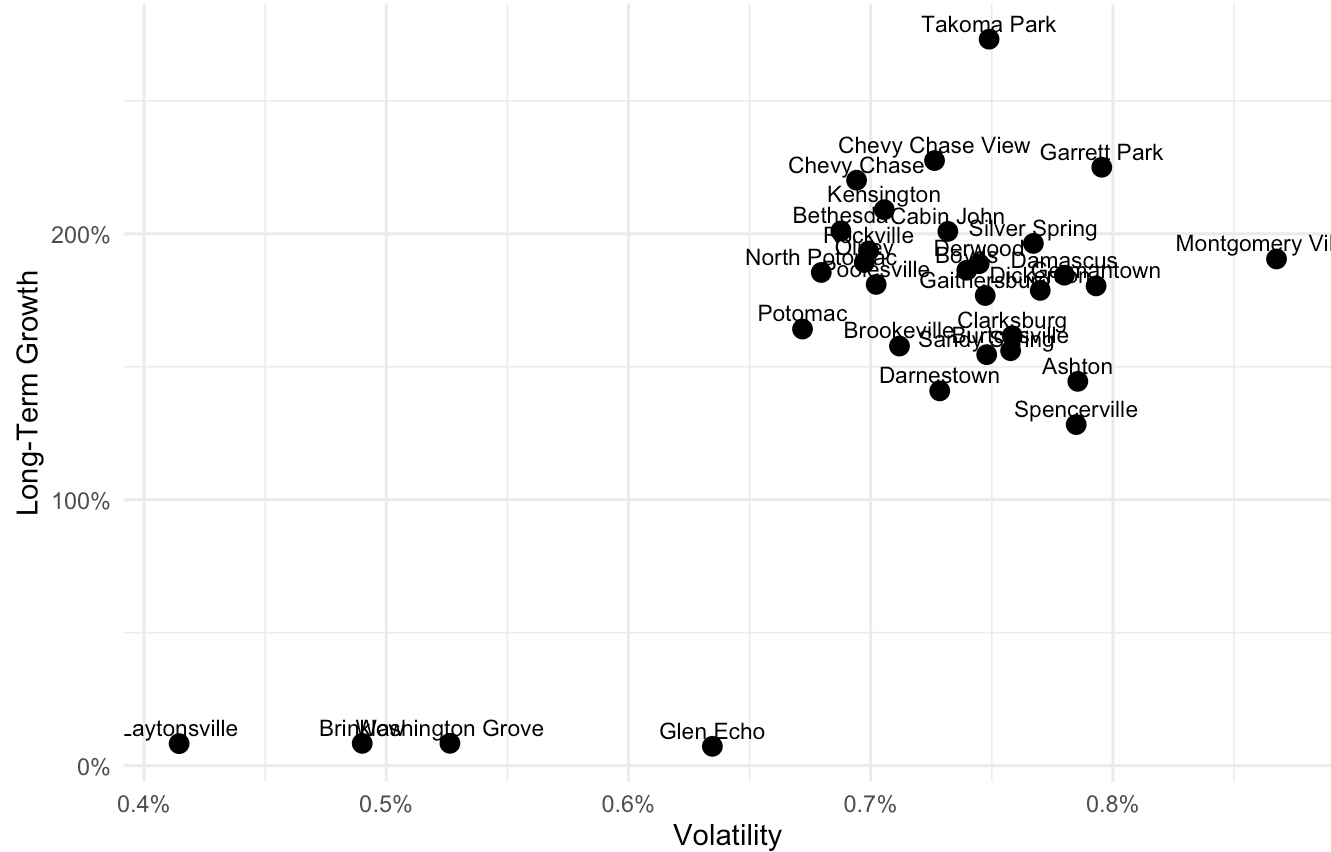
```
## # A tibble: 31 × 3
##   RegionName      avg_monthly_change volatility
##   <chr>          <dbl>          <dbl>
## 1 Montgomery Village      0.00344      0.00868
## 2 Garrett Park           0.00379      0.00795
## 3 Germantown             0.00332      0.00793
## 4 Ashton                 0.00288      0.00786
## 5 Spencerville           0.00266      0.00785
## 6 Damascus               0.00336      0.00780
## 7 Dickerson              0.00330      0.00770
## 8 Silver Spring          0.00350      0.00767
## 9 Clarksburg             0.00310      0.00758
## 10 Burtonsville          0.00303      0.00758
## # i 21 more rows
```

Growth vs Volatility

```
investment_summary <- growth_summary %>%
  left_join(volatility_summary, by = "RegionName")

ggplot(investment_summary, aes(x = volatility, y = growth_percent, label = RegionName))
+
  geom_point(size = 3) +
  geom_text(vjust = -0.5, size = 3) +
  scale_y_continuous(labels = percent_format()) +
  scale_x_continuous(labels = percent_format()) +
  labs(
    title = "Growth vs. Volatility in Montgomery County Housing Markets",
    x = "Volatility",
    y = "Long-Term Growth",
    caption = "Higher growth with lower volatility may indicate stronger investment potential."
  ) +
  theme_minimal()
```

Growth vs. Volatility in Montgomery County Housing Markets



Higher growth with lower volatility may indicate stronger investment potential.